Structural Analysis of the Global Material Crisis Following the 2026 Strait of Hormuz Blockade: Vulnerabilities and Ripple Effects of the Naphtha Supply Chain in Japan, the U.S., and China

1. The Flashpoint of the Crisis: Breakdown of the Islamabad Negotiations and the Severing of Global Supply Chains

The military conflict between the U.S./Israel and Iran that broke out in the spring of 2026 has transformed far beyond initial concerns of a Middle Eastern geopolitical crisis and energy price spikes, evolving into a "global material crisis" that strikes the physical foundations of modern civilization. The decisive turning point of this crisis was the collapse of direct negotiations between the U.S. and Iran—the highest-level talks since the 1979 Iranian Revolution—held in Islamabad, Pakistan, in April of the same year.

The U.S. delegation, led by Vice President J.D. Vance, along with U.S. Middle East Envoy Steve Witkoff and former Senior Advisor Jared Kushner, held a 21-hour top-secret discussion mediated by Pakistani Prime Minister Shehbaz Sharif.[1] The U.S. side demanded the complete and permanent abandonment of Iran's nuclear weapons development program and the unconditional reopening of the Strait of Hormuz, offering sanctions relief in return as their "final and best offer".[3] However, the Iranian side rejected this as containing "unreasonable demands," and countered by demanding recognition of its transit control rights over the Strait of Hormuz, reparations for past damages, and a region-wide ceasefire including Lebanon, resulting in a complete breakdown of the negotiations.[6]

Immediately after Vice President Vance stated, "We have not reached an agreement. I think that's bad news for Iran much more than it's bad news for the United States," and departed Islamabad, U.S. President Donald Trump declared a de facto naval blockade of the Strait of Hormuz.[9] President Trump announced a policy to seize and intercept in international waters any vessels passing through the strait that pay unjustified tolls to Iran, deploying U.S. Navy destroyers to the Persian Gulf.[12]

Prior to this heightening of military tensions, major marine insurance underwriting syndicates such as Gard and Skuld had already issued "Notices of Cancellation" for war risk insurance covering the Persian Gulf waters by early March.[14] Being excluded from insurance coverage

made the navigation of private merchant vessels legally and financially impossible, creating a structure where a "de facto complete naval blockade" was achieved even without physical destruction.[14] International Energy Agency (IEA) Executive Director Fatih Birol estimated that this situation erased approximately 11 million barrels of crude oil supply and about 140 billion cubic meters (BCM) of natural gas supply per day from the global market.[15] Director Birol declared this supply loss to be a complex shock unprecedented in history, surpassing the combined scale of the two oil shocks of the 1970s and the supply reduction caused by the 2022 Ukraine war.[15]

The dysfunction of the strait, through which approximately 20% of the world's seaborne crude oil transport (about 20 million barrels per day) passes, goes beyond soaring prices for fuels like gasoline and diesel. It has triggered a severe depletion of "naphtha" (crude gasoline), the fundamental raw material for the petrochemical industry refined from crude oil. This shortage is quietly but surely beginning to erode all aspects of daily life and high-tech industries, ranging from food packaging and medical devices to paints, textiles, and semiconductor-related materials.[14] This report strictly analyzes and delves into how the difference in the "quality" of crude oil imposes asymmetric impacts on the industrial structures of Japan, the U.S., and China, particularly concerning the supply chains for medical care, food, military, and daily necessities.

2. Structural Divergence in the Petrochemical Industry: Divisions Dictated by the "Quality" of Crude Oil

To accurately understand the impact of the naphtha shortage on each country, it is essential to grasp the technical differences between the molecular structural characteristics of crude oil in the upstream sector and the process designs of petrochemical plants in the downstream sector. The harsh reality that "not all crude oil yields basic chemicals at the same rate" fundamentally bifurcates the crisis resilience of Japan and the United States.

2.1 Middle Eastern Heavy Crude Oil and the Generation Mechanism of Aromatics (BTX)

When crude oil is refined (fractionated), it separates into LPG, gasoline, naphtha, kerosene, diesel, and heavy fuel oil based on boiling point differences. Middle Eastern crude oil generally consists of medium to heavy oils with an API gravity of 22 to 38 and features high sulfur content (sour crude). However, the heavy naphtha fraction obtained from this Middle Eastern crude oil is rich in naphthenic hydrocarbons with cyclic structures.

By processing this heavy naphtha through a refinery's catalytic reforming unit, a dehydrogenation reaction occurs, extracting "BTX (aromatic components)" consisting of Benzene, Toluene, and Xylene at a high yield.[18] Benzene becomes the raw material for polyurethane (MDI) and polycarbonate resins; toluene is used for TDI and various solvents; and

xylene (paraxylene) serves as the basic raw material for polyester fibers and PET resins. Furthermore, fluid catalytic cracking (FCC), generated during the refining of heavy crude oil, yields large quantities of propylene and butadiene, a raw material for synthetic rubber.[19] Modern advanced polymer industries and pharmaceutical synthesis are critically dependent on the aromatic components derived from this Middle Eastern crude oil.

2.2 Characteristics of U.S. Shale Oil and Structural Mismatch

In contrast, the shale oil (tight oil) that drove the U.S. to become the world's largest oil producer is a light paraffinic crude oil with high API gravity and extremely low sulfur content.[20] High-temperature thermal cracking of ethane (NGL), abundantly obtained from shale gas, in a steam cracker yields ethylene at an extremely high rate of up to 70–80%.[21] U.S. petrochemical manufacturers (such as Dow Chemical) operate these inexpensive ethane crackers at utilization rates exceeding 90%, establishing their position as the world's "swing producer" for the supply of general-purpose plastics (polyethylene) without relying on Middle Eastern naphtha.

However, the fatal weakness of ethane crackers is that they generate almost no fractions other than ethylene (such as propylene, butadiene, and aromatic components like BTX).[22] To compensate for this structural flaw, the U.S. has relied on refinery FCCs and imports for its lacking aromatic components and heavy fractions. The critical issue here is the historical structure of U.S. refinery infrastructure. From the 1980s to the early 2000s, approximately 70% of U.S. refineries made advanced capital investments in cokers (thermal cracking units) and hydrodesulfurization units to process "cheap heavy sour crude" from Canada, Mexico, the Middle East, and Venezuela.[23]

In other words, now that the supply of heavy crude oil from the Middle East has been cut off, the U.S. cannot maintain the optimal operation of its existing refineries using only the light shale oil gushing within its own borders. As a result, it faces a structural contradiction of falling into a devastating shortage of aromatic components, the lifeblood of the chemical industry.[23]

Table 1: Asymmetric Yields of Basic Chemicals by Raw Material Source

Raw Material Source	**Major Producing Regions**	**Processing Methods**	**High-Yield Basic Chemicals**	**Examples of Final Products**
Middle Eastern Heavy/Medium Crude	Middle East (Saudi Arabia, UAE, Iran, etc.)	Naphtha cracker, Catalytic reforming, FCC	**BTX (Benzene, Toluene, Xylene), Propylene,**	PET resins, Polyurethane, Synthetic rubber, Solvents, Active Pharmaceutical

			Butadiene	Ingredients (APIs), Paints
Shale Gas (Ethane)	USA	Ethane cracker	**Ethylene** (Propylene and BTX are extremely low)	Polyethylene (PE), Wrap films, General-purpose plastics
Coal	China	Coal liquefaction, Carbide process, CTO (Coal-to-Olefins)	**PVC, Basic olefins, Methanol**	Construction materials, Basic chemicals in general

3. Japan's Devastating Vulnerability: The Reality of Naphtha Depletion Concealed by "254 Days of Reserves"

The manifestation of the naphtha crisis in Japan exposes an inherent and fatal vulnerability distinct from both the U.S. and China. Japan is extremely dependent on overseas sources for the origins of its basic chemicals.

3.1 Naphtha Import Dependency and Declining Cracker Utilization

According to 2024 statistical data from the Japan Petrochemical Industry Association (JPCA), Japan relies on imports for approximately 60.6% of its domestic naphtha demand (about 2.03 million tons per month).[28] Furthermore, over 60% of that imported naphtha is supplied from the Middle East.[28] The blockade of the Strait of Hormuz is tantamount to completely severing the artery that forms the foundation of Japan's petrochemical industry.

The disruption of naphtha supply from the Middle East through the Strait of Hormuz (on a scale of 4 million tons per month on average) has triggered cascading operational declines across Asia.[28] The utilization rate of naphtha crackers in Northeast Asia fell to 73% in March.[29] A serious situation is also unfolding within Japan. Mitsubishi Chemical has reduced the utilization rate of its flagship naphtha crackers at the Kashima and Mizushima plants—with a combined ethylene production capacity of approximately 1.3 million tons per year—to 80–85%.[29] Idemitsu Kosan

has warned customers of potential equipment shutdowns at its Chiba and Yamaguchi plants, and crackers operated by Tosoh, Resonac (formerly Showa Denko), and ENEOS have been forced to postpone startups from scheduled maintenance or extend maintenance periods.[30] Out of the 10 crackers in Japan, at least 6 have been forced into production cuts or operational adjustments.[32]

3.2 Limitations of the National Stockpile Strategy and the Ripple Effects of an Additional "20-Day" Release

In response to this situation, the Japanese government was forced to take extraordinary measures. Following the decision in mid-March to release 15 days' worth of private reserves and 30 days' worth of national reserves, Prime Minister Sanae Takaichi announced a policy to release an additional 20 days' worth of national reserves (equivalent to approximately 11 million barrels) starting in early May.[33] As of the end of 2025, Japan boasted a world-leading stockpile of 254 days' worth (approximately 470 million barrels) in total, comprising national reserves (143–146 days), private reserves (101 days), and joint reserves with oil-producing countries (7 days).[33]

Prime Minister Takaichi has indicated a policy of weathering the immediate crisis by maintaining domestic refining levels at approximately 1.1 million kiloliters per month while doubling naphtha imports from non-Middle Eastern regions such as the U.S., Latin America, and Africa (to approximately 900,000 kiloliters per month).[28] However, the composition of the released crude oil reserves does not perfectly match the optimal operating conditions of Japan's refineries and crackers, and substitute naphtha from sources like the U.S. is poor in aromatic components. Consequently, domestic petrochemical manufacturers have fallen into a situation where they must pay abnormal premiums on the spot market to secure limited volumes.[28]

Table 2: Operating Status of Major Naphtha Crackers in Japan (As of March-April 2026)

Company Name	Major Plants	Estimated Capacity (Ethylene equivalent)	Operating Status and Response
Mitsubishi Chemical	Kashima, Mizushima	Approx. 1.3 million tons/year	Utilization rate reduced to 80-85%. Major customers notified. [29]
Idemitsu Kosan	Chiba, Yamaguchi	-	Warned of potential plant shutdowns.

			Supply constraint risks exist. [30]
Mitsui Chemicals	Chiba, Osaka	-	Seeking to secure naphtha from alternative non-Middle Eastern sources. [30]
ENEOS	-	-	Scheduled maintenance started in mid-March expected to be extended due to supply disruptions.
Tosoh / Resonac	-	-	Delayed restart from scheduled maintenance.

3.3 Direct Blows to Domestic Manufacturing: Supply Crisis of Paints and Packaging Materials

The decline in cracker utilization is immediately inflicting devastating impacts on downstream manufacturing. A survey by a paint industry association in the Kanto region revealed a situation where approximately 70% of companies could not secure the necessary raw materials in April.[40] The price of paint thinner, primarily made from toluene and xylene refined from the aromatic components of naphtha, showed an abnormal surge of 75% between March and April, with some becoming completely unobtainable.[36] The situation has become so severe that Prime Minister Takaichi had to directly order the resolution of bottlenecks for paint thinners used in housing construction and automobile maintenance at a cabinet meeting.[36]

Spillover effects into daily necessities are also inevitable. Due to surging raw material costs, price hikes of over 30% have been announced for everyday items such as garbage bags (made of polyethylene) and food packaging films starting in late May.[32] Product groups whose supply chains are completed domestically, such as polystyrene food trays and polyethylene wrap films, are expected to be barely maintained until the end of the year through priority allocation of stockpiled naphtha. However, PET resin (the raw material for PET bottles and retort pouches) manufactured from paraxylene is highly likely to face severe shortages within a few months due to the lack of aromatics compounded by a decrease in preform imports from China.

Japan's 254 days of reserves function as a device to "buy time" to prolong domestic production

networks, but they pose a structural limit in that they offer no fundamental solution for finished products manufactured overseas or high-performance chemicals requiring specific molecular structures (aromatics).

4. Ripple Effects on Medical Care and Public Health: Direct Threats to Life Due to Material Shortages

The most cruel and severe ripple effect caused by the naphtha shortage is the dysfunction of the healthcare system. Looking inside a hospital, nearly all consumables, from syringes to IV bags, are made of plastic. And without exception, their raw material is naphtha.

4.1 Depletion of Nitrile Gloves and Synthetic Rubber

Medical nitrile gloves, essential for surgery, treatment, and routine infection control, are manufactured from NBR (Nitrile Butadiene Rubber). NBR is a synthetic rubber copolymerized from butadiene, obtained during naphtha cracking, and acrylonitrile, obtained by the ammoxidation of propylene.[41]

In Malaysia, which supplies about 65% of the world's medical gloves (with domestic production accounting for about 45%), the Malaysian Rubber Glove Manufacturers Association (MARGMA) issued a statement on March 26 warning of a crisis situation: a "30% shortage in NBR supply".[41] Due to the halt in raw material supply from the Middle East caused by the blockade of the Strait of Hormuz, the price of NBR raw materials has doubled from $750 to $1,500 per ton.[41] Major manufacturers, including the world's largest, Top Glove, have been forced to postpone production plans and pass on significant price increases. MARGMA has urgently requested the Malaysian government to intervene by temporarily restricting overseas raw material exports and prioritizing supply to domestic glove makers.[41]

Japan relies on imports for nearly 100% of its medical gloves. Production delays and price surges in Malaysia translate immediately into a shortage of gloves in Japanese medical settings. Without gloves, surgeries, blood draws, and even responses to infectious diseases like COVID-19 will come to a complete halt. No matter how much of Japan's stockpiled crude oil is released, it cannot directly support glove production in Malaysia.

4.2 Risk of Hemodialysis Circuit Dysfunction: The Lifeline for 350,000 People

The hemodialysis system, requiring treatment three times a week for approximately 350,000 patients in Japan, is a culmination of advanced petrochemical polymers.[45] The rigid housing of the dialyzer is made from polycarbonate, synthesized from the aromatic component bisphenol A, and polypropylene.[46] The hollow fiber membranes inside, which filter waste products from the blood, are manufactured from highly advanced aromatic polymers such as polyethersulfone (PES) and polysulfone (PS).[47] Furthermore, the tubing that circulates blood outside the body uses medical-grade PVC (polyvinyl chloride) with added plasticizers (such as

DEHP).

All of these originate from BTX (aromatic components) extracted from heavy naphtha, as well as ethylene and propylene. As Takeshi Shukunobe, representative of a patient organization, appealed to the Ministry of Health, Labour and Welfare, "Dialysis is our lifeline, and even the rumor of a supply problem is tantamount to a death sentence".[48] The cessation of supply for irreplaceable advanced medical components directly threatens human lives on a different level than food shortages. The same applies to catheters (polyurethane, silicone) and IV/blood bags (PVC).

If it were food, the crisis might be mitigated by reallocating domestic agricultural production. However, hemodialysis circuits and cardiac catheters cannot be harvested from fields. "People dying because of plastic problems"—this is the harshest reality brought about by the material crisis.

5. Active Pharmaceutical Ingredients (APIs) and KSMs: The "Hollow Shelf" Risk Controlled by China

Beyond plastic medical devices, pharmaceuticals themselves (tablet coatings, capsules, solvents) are products of petrochemistry. The greatest risk here is the supply network of Key Starting Materials (KSMs) essential for synthesizing APIs (Active Pharmaceutical Ingredients), the active components of medicines.

5.1 Structural Vulnerabilities Revealed by the USP Medicine Supply Map

Recent data analysis from the United States Pharmacopeia (USP) "Medicine Supply Map" highlights a fatal vulnerability in the medical security of Japan and the U.S. According to the data, a staggering 58% of KSMs used in manufacturing APIs approved in the U.S. rely on single-country sourcing. Even more alarmingly, 41% are dependent on sole-source supply from China, and 16% on sole-source supply from India.[50]

While India possesses approximately 48% of the world's API manufacturing sites (based on DMF registrations for the U.S.) and serves as the global supply base for generic drugs, a dual structure exists where India's API industry itself completely relies on imports from China for 70–74% of its KSMs and bulk chemicals.[53] Aromatic components like benzene and toluene are fundamental KSMs for the chemical synthesis of antibiotics (such as Penicillin G) and antipyretic analgesics (such as paracetamol).

5.2 The Advent of the "Hollow Shelf" Phenomenon

As the supply of heavy naphtha and aromatic components from the Middle East to Asia is cut off by the blockade of the Strait of Hormuz, Chinese chemical plants are forced to reduce KSM production. If Indian API factories consequently halt operations, the "Hollow Shelf" phenomenon—where basic antibiotics, anesthetics, and chronic disease medications vanish

from hospitals and pharmacies in Japan and the U.S.—will become a reality within months.[53]

Both Japan and the U.S. possess several months' worth of stockpiles for finished pharmaceuticals. However, because the exporting countries themselves will be unable to produce due to a lack of raw materials (KSMs), domestic stockpile strategies alone cannot save the system. Capitalizing on the fact that it controls 41% of the global KSM market, China is seizing pricing power amidst this supply shortage, further strengthening its geopolitical leverage.

Table 3: Vulnerability Map of Medical and Pharmaceutical Supply Chains

Medical Item	Primary Material / Chemical Raw Material	Dependent Upstream Naphtha Fraction	Global Supply Chain Bottleneck	Critical Impact
Nitrile Gloves	NBR (Acrylonitrile + Butadiene)	Propylene, Butadiene	Malaysia (65% global share) facing a 30% raw material shortage. [41]	Halt of surgical procedures and infection control. Prices have already doubled.
Hemodialysis Circuits	Polycarbonate, PES, PVC	Aromatics (Benzene), Ethylene	Directly linked to the life support of 350,000 patients in Japan.	Depletion of irreplaceable life-support equipment.
Active Pharmaceutical Ingredients (APIs)	KSMs (Key Starting Materials)	Aromatics (Benzene, Toluene, etc.)	China monopolizes 41% of global KSMs. Indian API plants face shutdown risks. [50]	"Hollow shelf" of generic drugs such as antibiotics.
Syringes / Masks	Polypropylene (including	Propylene	Naphtha shortages at	Stagnation of basic medical

	non-woven fabrics)		overseas manufacturing hubs like China.	procedures.

6. The U.S. Faces a Self-Contradiction: Blind Spots in the Defense Industry and the Depletion of Advanced Materials

Through the shale revolution, the U.S. achieved energy independence, becoming the world's largest oil and gas producer. Petrochemical manufacturers, led by Dow Chemical, operate ethane crackers at over 90% utilization rates, declaring they can maintain the supply of general-purpose plastics (polyethylene, etc.) "at full capacity until the end of the year." However, this strength has inverted into its greatest weakness. The deficiency of aromatic components (BTX) is causing a severe self-contradiction within the defense industry, the foundation of U.S. national security.

6.1 The Mismatch Between Shale Oil and Refineries

As mentioned earlier, about 70% of U.S. refinery capacity relies on cokers and heavy oil cracking units designed to process heavy sour crude from Canada, Mexico, and the Middle East.[23] Shale oil is extremely light and suitable for producing gasoline and ethane, but overwhelmingly lacks the heavy bottom fractions that should be fed into catalytic reformers and FCCs to produce aromatics. As a result, the U.S. cannot maintain optimal refinery operations through its domestic crude oil production increases alone, falling into a severe shortage of aromatic components and propylene.[23]

6.2 Depletion of Advanced Materials Supporting U.S. Military Warfighting Capabilities

This shortage of aromatics and propylene strikes directly at the U.S. defense industry. The latest U.S. military weapons, aircraft (such as the F-35), unmanned aerial vehicles (drones), and soldier protective gear rely on extremely advanced petrochemical materials.

For instance, Carbon Fiber Reinforced Plastics (CFRP), essential for lightening and strengthening aircraft and missiles, requires PAN (polyacrylonitrile) as its precursor. Acrylonitrile, the main raw material for PAN, is synthesized through the ammoxidation reaction of propylene and ammonia.[55] Furthermore, aramid fibers (like Kevlar) used in bulletproof vests and helmets that protect soldiers' lives are made from aromatic polyamides, requiring BTX components like toluene and benzene.[57]

In other words, the naval blockade of the Strait of Hormuz—spearheaded by the Trump administration against Iran—is rebounding on Washington as a fatal self-contradiction (a

boomerang effect), independently depleting the raw materials for the advanced materials that support the U.S. military's own warfighting capabilities. Amidst reigniting inflation and soaring gasoline prices shaking domestic support bases, the U.S. is caught in a dilemma where the "materials" needed to sustain a war are lost because of its own war.

7. China's Asymmetric Advantage and New Achilles' Heel

In this unprecedented global material crisis, China holds the most geopolitically and industrially advantageous position. However, China is not completely unscathed and simultaneously harbors vulnerabilities as a massive energy consumer.

7.1 China's Strengths: The World's Largest Coal Chemical Industry and Massive Reserves

China's greatest bulwark is its "coal chemical industry," built on a scale unparalleled in the world. According to data from the China Petroleum and Chemical Industry Federation (CPCIF), as of 2024, China's modern coal chemical sector (coal liquefaction, CTG: coal-to-gas, CTO: coal-to-olefins, CTEG: coal-to-ethylene glycol) possesses a massive processing capacity of 138 million tons per year (standard coal equivalent).[58] This allows China to synthesize PVC (via the carbide method) and basic olefins using abundant domestic coal, without relying on Middle Eastern crude oil or imported naphtha, maintaining an operating rate of about 80%.

Furthermore, China has been aggressively building up strategic petroleum reserves over the past few years. According to estimates by oil market analysis firms Vortexa and Kpler, China stores between 1.39 billion and 1.47 billion barrels of crude oil onshore, equivalent to 104 to up to 130 days of imports.[59] Additionally, stable crude oil supply through pipelines from Russia (such as ESPO) partially hedges the risks of a maritime blockade.

7.2 China's Achilles' Heel: ZPC Production Cuts and Reliance on Mega-Infrastructure

On the other hand, China's greatest vulnerability lies in the sheer scale of its imports. According to Vortexa data, a staggering 37.7% of all crude oil passing through the Strait of Hormuz is destined for China, representing the largest share of any single nation globally.[59]

This vulnerability became most apparent with the operational adjustments at Zhejiang Petrochemical Corp (ZPC), China's largest single-site refining and petrochemical complex. ZPC relies on Saudi Arabia for about 60% of its crude oil requirements (about 480,000 barrels per day) under a long-term contract with Saudi Aramco, and when factoring in Middle Eastern crude oil via other routes, its dependence on crude oil passing through the Strait of Hormuz is overwhelming.[60] The rigidity of long-term contracts made shifting to alternative procurement difficult, forcing ZPC to bring forward maintenance on a 200,000-barrel-per-day crude distillation unit (CDU) and reduce throughput by 20%.[61]

Moreover, a joint venture between Shell and CNOOC (China National Offshore Oil Corp) indefinitely halted its ethylene cracker (1.2 million tons/year) in Huizhou, Guangdong Province, due to a lack of raw materials, suspending polyethylene shipments.[62] Wanhua Chemical, a global manufacturer of polyurethane raw materials (MDI/TDI), also declared force majeure on product supplies to the Middle East.[64]

Furthermore, China is the world's largest importer of methanol, and about 33% of the seaborne methanol trade passes through the Strait of Hormuz.[66] According to analyses by the World Economic Forum (WEF) and others, methanol inventories are approaching warning levels. Additionally, supply shortages of urea (the main raw material for fertilizers, with 46% of global seaborne trade passing through the strait) and sulfur (about 50% of global trade depends on this route), which becomes the raw material for sulfuric acid indispensable for EV battery manufacturing (copper, nickel, cobalt refining), harbor the potential to become growth Achilles' heels for China's agricultural and high-tech industries.[66]

Table 4: Dependence on the Strait of Hormuz for Major Commodities and the Impact

Commodity	**% of Global Seaborne Trade via Strait of Hormuz**	**Main Applications / Downstream Industries**	**Most Impacted Countries / Regions**
Crude Oil	Approx. 20%	Gasoline, Naphtha, Heavy fuel oil, All petrochemicals	China (Receives 37.7% of total transit volume), Japan, India
Sulfur	Approx. 50%	Sulfuric acid production, Fertilizers, EV battery metal refining	China (EV industry), Global agricultural sector
Urea (Ammonia-based)	Approx. 46%	Nitrogen-based fertilizers, Adhesives	Global agricultural sector (Trigger for food crisis)

Methanol	Approx. 33%	Formaldehyde, Resins, Paints, Synthetic fibers	China (World's largest importer), All of Asia
LNG (Liquefied Natural Gas)	Approx. 19-20%	Power generation, City gas, Certain chemical feedstocks	All of Asia (Japan, South Korea, Taiwan, China)

7.3 "Profiting from the Conflict": The Completion of Material Hegemony

Despite harboring vulnerabilities, in relative terms, China holds the most advantageous position in this crisis. While petrochemical manufacturers in Japan, the U.S., and Europe lower their operating rates, China can sustain basic chemical production through coal chemistry and keep refineries partially operational utilizing its reserves of over 104 days and Russian crude oil.

The longer the blockade of the Strait of Hormuz drags on, the more the "surviving nodes" in global supply chains will concentrate in China. The structure is irreversibly advancing through this crisis: the "departure from fossil fuel dependence" promoted by Western nations is ironically being supplanted by an "absolute dependence on Chinese materials" (such as KSMs, basic chemicals, and alternative materials).

8. Recovery Timeline and the "Circular Restoration Problem"

Even if negotiations in Islamabad were to dramatically resume and the U.S. and Iran compromised to lift the blockade of the Strait of Hormuz, global supply chains would not normalize overnight. A massive physical and chemical barrier known as the "Circular Restoration Problem" stands in the way.

8.1 Physical Destruction of Gulf Infrastructure

The massive energy infrastructure in the Gulf region has already sustained severe physical damage from Israeli/U.S. airstrikes on Iran and Iran's retaliatory exchanges. According to Roland Lescure, French Minister Delegate for Industry, Iranian drone strikes damaged Qatar's Ras Laffan industrial zone (a downstream product complex for LNG, polymers, methanol, urea, etc.), wiping out approximately 17% of Qatar's gas production and export capacity. Lescure points out that "the complete restoration of these facilities will require about three years".[69] Additionally, a joint refining facility with TotalEnergies on the eastern coast of Saudi Arabia

(SATORP) has halted operations following multiple attacks.[12]

8.2 The Circular Restoration Problem: Factories That Cannot Be Fixed

Repairing and rebuilding destroyed colossal petrochemical plants and pipelines requires vast amounts of advanced materials, such as anti-corrosion coatings, special sealants, polyurethane insulation, industrial paints, and strong adhesives. However, all of these materials are "naphtha-derived petrochemical products."

In short, because the very mega-facilities that should be supplying naphtha and basic chemicals have been destroyed, the world is trapped in a deadlock: "There are no materials to fix the factories that make the materials needed for restoration." Crude oil can be transported across oceans as long as tankers are secured, but a destroyed distillation tower or ethylene cracker standing tens of meters tall cannot be instantly shipped by boat. Market expectations that "everything will be resolved once the Strait of Hormuz opens" are nothing but illusions.

Table 5: Projected Timeline for Supply Chain Recovery Post-Blockade

Recovery Phase	Estimated Duration	Recovery Bottlenecks and Constraining Factors
Resumption of Seaborne Crude Oil Transport	2 to 4 weeks	Mine-sweeping operations, Re-contracting marine insurance, and normalization of premiums.
General-Purpose Plastics (PE/PP, etc.)	3 to 6 months	Process of restarting Asian crackers. Rebuilding inventories.
Aromatics-derived Products (PET, Synthetic Rubber, etc.)	6 to 12 months+	Dependent on the progress of restoring destroyed Middle Eastern catalytic reformers and refineries. Material shortages due to the "Circular Restoration Problem."

Active Pharmaceutical Ingredients (APIs/KSMs)	6 months to 3 years	Restart of Indian factories and restructuring procurement routes from China. Time lag in resolving "Hollow Shelves."
Mega-Plants (Methanol, Urea, LNG)	1 to 3 years	Physical reconstruction of massive complexes in Qatar, etc. Period of spillover into the global food crisis.

9. Conclusion: Paradigm Shift and Strategic Recommendations

The essence of the "Naphtha Crisis," highlighted by the 2026 Iran War and the ensuing blockade of the Strait of Hormuz, is the stark reality of how fragilely modern civilization rests upon the monoculture of "petrochemistry." From food packaging, medical devices, and pharmaceuticals to clothing, construction materials, agricultural fertilizers, and even cutting-edge military defense capabilities—the roots of all industries converged on a single geographical chokepoint: the Strait of Hormuz.

Japan's strategic stockpile of "254 days and 470 million barrels" served as an effective buffer to suppress gasoline prices and prolong general-purpose resins completed domestically. However, it was virtually powerless against "materials" heavily dependent on overseas supply chain sources, such as medical nitrile gloves, hemodialysis circuits, and active pharmaceutical ingredients (KSMs). The U.S. achieved energy independence through the shale revolution, yet exposed a fatal self-contradiction: due to the lightness of its crude oil and the resulting lack of aromatic components, it cannot self-supply the advanced materials necessary to sustain its military power and advanced medical care.

Conversely, China has relatively minimized damage by leveraging its unique buffer of the coal chemical industry and massive strategic reserves. Consequently, an ironic reversal is underway, deepening the dependency of Japan, the U.S., and Europe on Chinese materials (KSMs and basic chemicals).

As decarbonization and the energy transition are globally promoted, the international community has been paving a path toward moving away from "oil as a fuel." However, this crisis has demonstrated just how difficult it is to transition away from "oil as a material (Petrochemicals)," and how the severing of this supply immediately threatens human lives and national survival.

Japan, the U.S., and other nations face the urgent task of making the following strategic shifts towards post-crisis stabilization. First, thoroughly implement "friend-shoring" to repatriate manufacturing bases for Key Starting Materials (KSMs) and medical polymers within allied spheres. The extreme reliance on specific countries (such as China's 41% monopoly) revealed by USP's Medicine Supply Map must be rectified as a paramount national security flaw. Second, build "independent and circular material supply chains" that do not rely on specific geographical chokepoints or particular "qualities" of crude oil, by socially implementing chemical recycling technologies from waste plastics (such as re-extracting BTX from waste plastics) and bio-based materials.

As the collapse of the Islamabad negotiations showed, geopolitical risks are entering a realm beyond control. Even if the blockade were lifted tomorrow, restoring lost supply chains would take years. Even at this very moment, in the depths of medical sites and manufacturing industries, the "collapse of materials" is quietly, yet inexorably, progressing.

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